

# Adaptive Federated Reinforcement Learning With Temporal Hybrid Deep Model for Consumer Internet of Vehicles Intrusion Detection

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**Abstract**—Consumer devices that are integrated into the Internet of Vehicles (IoV) systems are becoming more susceptible to the sophisticated intrusion attacks, which are a major threat to user safety and the overall integrity of the system. Federated Learning (FL) has been a prominent technique for collaborative training models while maintaining the confidentiality of sensitive information. In addition, conventional FL-based IDS approaches have problems to handle non-identical IID (IID) data distributions across clients, and usually use a fixed aggregation method like FedAvg and FedProx, which has poor adaptability to changeable client behaviors. To tackle these limitations, this research proposes an adaptive federated learning framework by preprocessing IoVs network traffic through standardization and IID (independent and identical distributed) distribution of data among clients. A Temporal Hybrid Deep Model (CNN-LSTM-Attention) is trained on each client locally (starting from the global model). After training, client models and performance metrics are updated to a central server, where a Reinforcement Learning (RL) agent based on Deep Q-Network (DQN) optimally decides aggregation weights based on client Q-values, which maximizes global accuracy as a reward. Iterative aggregation greatly improves intrusion detection performance, with detection accuracies of 99.85% and 95.18% on the benchmark datasets CICIOV2024 and CICEVSE2024, respectively. The proposed method makes it possible to develop a lightweight, edge-deployable intrusion detection system for consumer IoV device like telematics and electric car chargers. By leveraging federated learning, the approach provides real-time, privacy-preserving protection.

**Index Terms**—Federated learning, Internet of Vehicles, intrusion detection system, reinforcement learning, cybersecurity.

## I. INTRODUCTION

CONSUMER-ORIENTED cybersecurity attacks on the Internet of Vehicles (IoV) have become a growing concern as IoV technologies are improving and increasingly filling the day-to-day lives of individuals in attempts to create greater convenience, safety and driving experience to the end user. Recent studies have shown that the threats to consumers are increasing. These include data breaches, unauthorized access, spoofing, denial of service (DoS) and ransomware, all of which directly damage consumer trust and lower the adoption of IoV. These threats demonstrate how important it is to have strong security measures in place to protect consumers as well as ensure sure that IoV applications keep on growing.

The Internet of Vehicles (IoV) is an integral component of the Internet of Things (IoT), and is causing a revolution in transportation by allowing vehicles, infrastructure and personal gadgets to connect to each other in new ways. Advanced sensor technologies built into cars and roadside devices now generate an extensive amount of data, which is supporting innovative applications which make transportation more efficient and improve the user experience [1]. Both government and industry leaders all over the world are advocating for the use of intelligent transportation systems (ITS) fueled by IoV. These systems can be used for real time traffic monitoring, self-driving cars, automated toll collection and comprehensive vehicle to everything (V2X) communications [2], [3], [4]. The consumer plays an important role in this scenario as the success and the social impact of IoV rely heavily on trust and engagement by the end user.

The deployment of fifth generation (5G) cellular networks has accelerated the possibilities of IoV further by adding greater reliability, higher throughput, lower latency and ubiquitous connectivity, which are necessary for smooth and real time IoV services [5], [6]. These technological advances have increased the Internet of Things (IoT) reach, and it has extended its reach to other areas like smart homes, health monitoring, and, importantly, consumer-based Intelligent Transportation Systems (ITS) solutions.

One of the things that makes IoV stand out is that it allows several different types of communications, including the following: vehicle-to-vehicle (V2V), vehicle-to-infrastructure

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(V2I), vehicle-to-pedestrian (V2P), and vehicle-to-network (V2N) interactions. These modes of communication work in combination with each other to make things safer, more efficient and comfortable for users. They often use advanced machine learning and artificial intelligence to improve service delivery.

As attacks on consumers become more sophisticated, modern research focuses on creating robust authentication protocols, secure communication frameworks, and smart intrusion detection systems that deal with the unique issues that consumer IoV environments face. Privacy, security and data integrity are important to ensure consumer confidence to ensure the full benefits of IoV adoption for society.

Two important problems of federated learning in 5G enabled Consumer Internet of Vehicles (IoV) environment such as non-identically and independently distributed (Non-IID) data across client devices because of which model performance and convergence problem occur [7]. Additionally, static aggregation methods like FedAvg and FedProx are not highly adaptable to variability in client capabilities and network dynamics and thus, they fail to be effective in real life consumer-centric IoV applications [8]. This study has the following contributions for solving the challenges that exist in current federated learning solutions:

- A federated learning framework is proposed for detecting malicious activity in 5G-enabled Consumer Internet of Vehicles (IoV). It is designed specifically to deal with non-IID data distribution and overcome the problems with static aggregating approaches like FedAvg and FedProx.
- The framework applies a Temporal Hybrid Deep Model (CNN-LSTM-Attention) to train each client locally, initializing the weights from the global model. A Deep Q-Network (DQN) agent based on reinforcement learning is additionally employed at the server to dynamically assign aggregation weights to each client based on their Q-values, which optimizes global model updates.
- Extensive experiments on the CICIoV2024 [9] and CICEVSE2024 [10] attack detection datasets validate the framework's effectiveness, with findings exhibiting superior performance compared to typical federated learning baselines in IoV intrusion detection scenarios.

The structure of the paper is as follows: Section II reviews related work, Section III explains the proposed method, Section IV discusses about the outcomes, and Section V concludes.

## II. RELATED WORK

The Internet of Things (IoT) industry is growing quickly, and this shift is closely related to the adoption of 6G networks, resulting in security challenges more severe [11]. IoT technologies have significantly transformed the networking and communication landscape [12]. Generally, definitions of the Internet of Things (IoT) and the Internet of automobiles (IoV) refer to physical entities that are connected to each other, including sensors and automobiles. However, there can be different interpretations [13]. The concept of IoT/IoV now covers everything from local networks to large networks, such

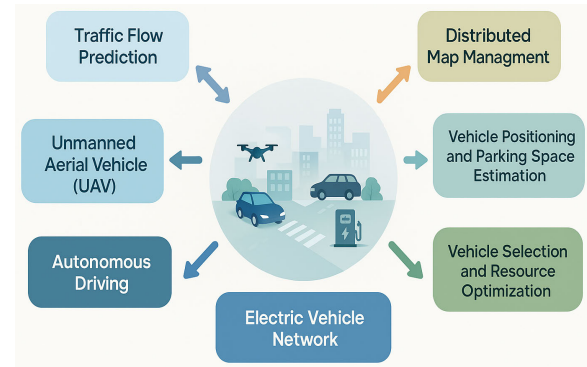


Fig. 1. Applications of Federated Learning (FL) in Internet of Vehicles (IoVs).

as LANs and the emerging 6G infrastructure. This change is mainly because 5G can't handle the high demands from emerging IoT and IoV applications [14]. As more devices connect to the Internet of Things (IoT) and Internet of Vehicles (IoV) systems, the amount of network traffic continues to expand. These technologies enhance daily life and industrial processes better, but they are also witnessing more malware threats at all the levels and throughout the entire network's life cycle [15]. Since it was first introduced, federated learning has drawn a lot of attention for its ability to solve problems related to data privacy, communication overhead, and client data variability [16]. Federated reinforcement learning (FRL) is a method that combines edge computing with distributed deep learning. Its aim is to enable urban sensing systems run more efficiently [17]. Recent studies show that decentralized data stored on devices may assist in identify IoT-related threats directly at the edge layer [18]. This approach applies Federated Learning (FL) as its main method, therefore there is no need for moving sensitive data offsite for processing [19]. FL not only speeds up model training on edge devices by keeping data local, but it also greatly improves data security. The emergence of the fourth industrial revolution and progress in communication technologies have driven the adoption of digital twins (DT), federated learning (FL), and the Industrial Internet of Things (IIoT). However, implementing DT and FL in IoV remains challenging. Recent studies have examined applications of IoT, IoV, and the Internet of Drones (IoD) for digital transformation (DT) and federated learning (FL) [20].

Federated learning (FL) has been extensively explored for improving Internet of Vehicles (IoV) applications. Previous research has examined FL across various fields, including to Autonomous Aerial Vehicles (AAVs), autonomous driving systems, and electric automobile networks [21]. Additional research has addressed areas like vehicle selection, resource optimization, and traffic flow prediction. Other topics include map management, content caching, vehicle positioning, and estimating parking spaces [22], as illustrated in Fig. 1. Vehicles are typically unwilling to send their raw local data straight to roadside units (RSUs) because they have concerns about privacy and security. In this situation, federated learning (FL) is a good way to do vehicular edge computing (VEC) since it lets cars communicate only their model parameters with adjacent RSUs instead of sensitive local data. This approach

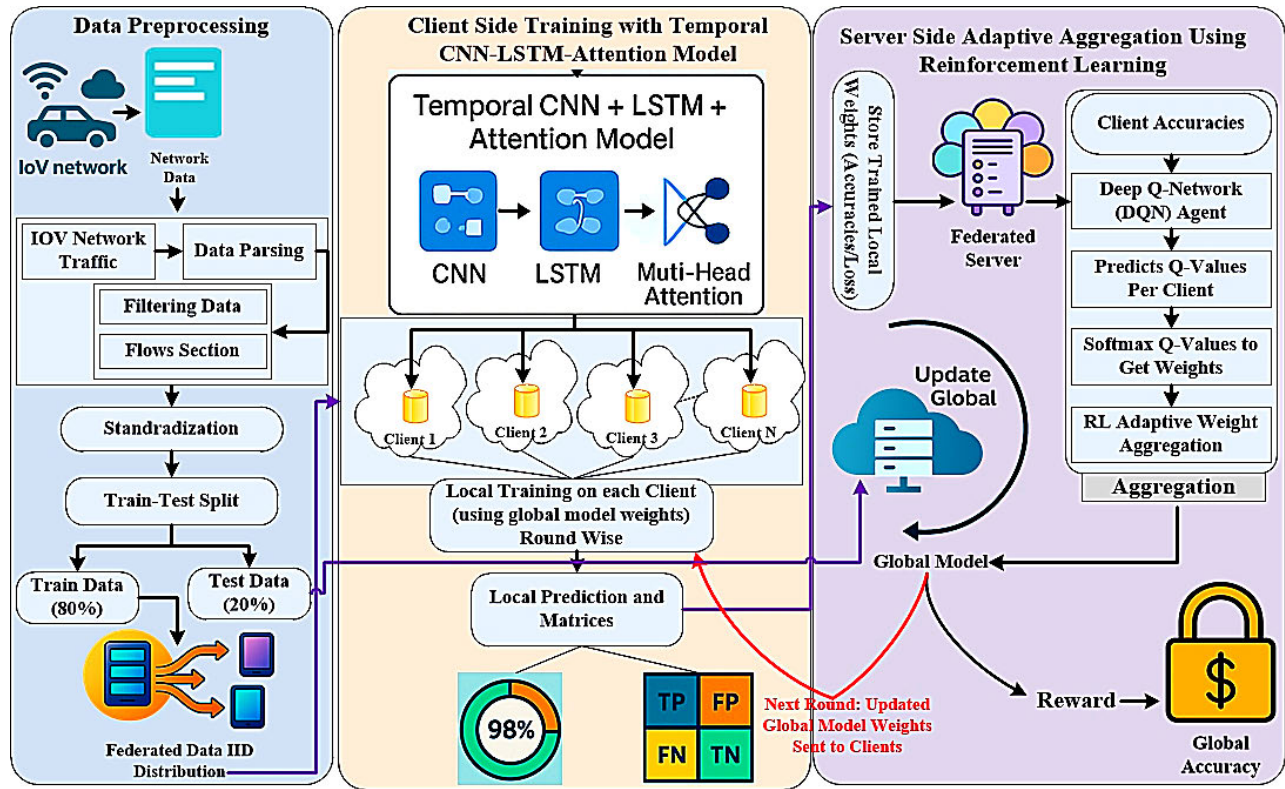


Fig. 2. Adaptive federated learning for 5G-enabled consumer internet of vehicles intrusion detection.

makes FL a good choice for machine learning problems in vehicular networks [23]. Collaboration among Vehicle-to-Everything (V2X) nodes is essential for building effective machine learning-based attack detection models in 5G and beyond vehicular networks [24].

Federated learning in IoV environments faces various challenges, like non-IID data distribution among clients, limited adaptability of static aggregation methods and vulnerability to underperforming or malicious participants. Additional challenges involve making it scalable as the number of clients and data increase and maintaining computational and communication efficiency. These things can make it so that the models do not converge as easily, are not accurate, and are not as strong in real life scenarios. Our proposed solution solves these problems by adopting a reinforcement learning-based (DQN) adaptive aggregation technique that assigns different weights to clients based on their performance. This method lowers communication overhead while increasing accuracy, robustness, and scalability when combined with multi-head attention and a hybrid CNN-LSTM model.

### III. ADAPTIVE FEDERATED LEARNING FOR 5G- ENABLED CONSUMER INTERNET OF VEHICLES INTRUSION DETECTION

This work tackles two major issues in the context of federated learning for 5G enabled Consumer Internet of Vehicles (IoV) networks namely: 1) non-IID data distribution among clients and 2) the lack of adaptability of the static aggregation techniques like FedAvx and FedProx. The proposed framework preprocesses IoV network traffic, standardizes and

splits the data and distributes it IID-wise among clients. Each client learns a Temporal Hybrid Deep Model (CNN-LSTM-Attention), based on its local data, with global weights. After each round, the clients send their trained weights and performance metrics to the server and a Deep Q-Network (DQN) agent is used to adaptively assign the aggregation weights based on client Q values. The softmax normalized weights update the global model which is re distributed to clients for the next round. This iterative process using global accuracy as a reward optimizes aggregation and intrusion detection performance strength for Internet of Vehicles. Fig. 2 shows the proposed adaptive federated learning framework for secure IoV intrusion detection based on reinforcement learning aggregation.

#### A. Data Preprocessing

The data preprocessing stage is important to ensure that the raw IoV network data is clean, structured, and suitable for intrusion detection using deep learning models. The first step is to gather the raw network information from the Internet of Vehicles (IoV) environment, which includes information about communication between vehicles, roadside units and IoT infrastructure. This raw data, typically in the form of packet capture (PCAP) files or logs, is parsed to extract useful features and converted to a structured feature matrix that can be used for analysis.

After the parsing is done, there might be irrelevant or noisy data in the dataset, such as incomplete data or duplicate flows or corrupted packets. The unwanted items are filtered in order to obtain a clean dataset. The network flows are then created

from the cleaned data, where the flows aggregate packets or events with common properties like source or destination IP, protocol, session or time window.

Next, the mean and the standard deviation of each feature in the dataset is scaled to have a mean of zero and a standard deviation of one, an important step in effective neural network training. For each feature vector  $x$ , standardization is carried out as follows:

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

where  $\mu$  and  $\sigma$  are the mean and standard deviation of the feature respectively. This is performed for each feature dimension and we finish with a standardized feature matrix. The standardized dataset is then divided into subsets for training and testing so that it allows for proper validation of the model and prevents overfitting, with usually 80% of the data being used for training and 20% being used for testing. After these preprocessing steps, processed data is then ready for IID data distribution between federated learning clients.

#### B. Client-Level IID Data Partitioning in Federated Learning

One of the main challenges of Federated Learning (FL) is the availability of non-identically and independently distributed (non-IID) data across the participating clients. In real world scenarios in FL, each client collects information from its own individual environment, leading to a lot of diversity in local data distributions and the risk of divergence in the local models. This data heterogeneity can significantly reduce the effectiveness of federated aggregation strategies and hinder overall model performance, especially in critical applications such as the Internet of Vehicles (IoV). To overcome this, the proposed framework includes data preprocessing that ensures the allocation of client-level IID (independent and identically distributed) data in order to have a more balanced and representative data set for each client's training locally. The workflow is random shuffling of indices of the whole data and equally dividing the data between the clients. Specifically, given a dataset  $D$  with  $|D|$  samples for  $N$  clients then number of samples per clients is defined as

$$S = \left\lfloor \frac{|D|}{N} \right\rfloor, \quad (2)$$

where  $S$  is the number of samples allocated in each client. After generating a random permutation  $\pi$  of the dataset indices, each client  $i$  is allocated a segment given by

$$D_i = D_{\pi(j)} \mid j = iS, \dots, (i+1)S - 1, \quad (3)$$

Ensuring that the subset of data is similar to the global distribution across all clients in order to achieve a realistic simulation of an IID setting for federated training. By enhancing uniformity of the client level data, this enhances greater stability and fairness of the global aggregation of the model, reduces inter client bias, and supports a better evaluation of federated learning algorithms.

#### C. Client Side Temporal Hybrid Deep Model Training

In the proposed federated learning framework for secure intrusion detection in IoV environments, the client side training is an important step in the process which leverages a powerful deep learning model allowing to learn local data patterns without the loss of privacy. Once data preprocessing and IID distribution are completed, each client node is given a divided data set as well as a copy of the global model weights which ensures the training begins with a standard initialization. Each client then starts local training based on a Temporal Hybrid Deep Model of a Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM) Network and Multi-Head Attention mechanism. The major goal of this stage is to optimize the local model parameters,  $\theta_k$ , for each client  $k$ , by minimizing the loss function using the client's local dataset  $(X_k, y_k)$ :

$$\theta_k^* = \arg \min_{\theta} \frac{1}{|X_k|} \sum_{i=1}^{|X_k|} \mathcal{L}(f_{\theta}(x_i), y_i) \quad (4)$$

Here,  $f_{\theta}$  is the deep model, and  $\mathcal{L}$  is the classification cross-entropy loss. Stochastic gradient descent (SGD) or an adaptive optimizer is applied each time to update the model of each client during training. After each round of communication, the local prediction results, e.g. accuracy, precision, recall, confusion matrix elements are calculated and reported. At the completion of local training, the updated model weights and local performance metrics are sent back to the federated server. This process is repeated for a number of rounds, with each round starting from the updated global model, and therefore, each round improving the ability of the model to generalize over all the data of the clients.

The Temporal Hybrid Deep Model is a composite model that tries to extract and combine short term and long term temporal dependencies within IoV network data, which is often highly sequential and structured. The model takes as input the sequence  $X \in \mathbb{R}^{s \times d}$ , where  $s$  is the length of the sequence and  $d$  is the dimension of features.

1) *TCN Pathway*: One branch goes through stacked 1D convolutional layers (Conv1D), each one has kernel size 3 and ReLU activation, which is responsible for capturing local temporal patterns:

$$\mathbf{h}_{\text{conv}}^{(l)} = \text{ReLU}(\text{Conv1D}(\mathbf{h}_{\text{conv}}^{(l-1)}, W_{\text{conv}}^{(l)})) \quad (5)$$

where  $l$  indexes the convolutional layer. Dropout is applied for regularization. A Global Average Pooling layer follows, aggregating features across time:

$$\mathbf{h}_{\text{tcn}} = \frac{1}{s} \sum_{t=1}^s \mathbf{h}_{\text{conv}}^{(L)}[t] \quad (6)$$

where  $L$  is the last convolutional layer.

2) *LSTM Pathway*: Simultaneously, the input sequence is fed into an LSTM layer, which models long-term dependencies:

$$\mathbf{h}_{\text{lstn}} = \text{LSTM}(X) \quad (7)$$

A dense layer is used to further transform the LSTM output.

3) *Feature Concatenation and Attention*: The outputs of the TCN and LSTM paths are concatenated:

$$\mathbf{h}_{\text{concat}} = \text{Concat}([\mathbf{h}_{\text{tcn}}, \mathbf{h}_{\text{lstm}}]) \quad (8)$$

This combined feature vector is reshaped as required and passed through a Multi-Head Attention mechanism, which learns interdependencies between temporal features:

$$\mathbf{h}_{\text{attn}} = \text{MultiHeadAttn}(\mathbf{h}_{\text{concat}}) \quad (9)$$

4) *Classification Head*: The attention output is flattened and passed through one or more dense layers with ReLU activation to produce the final class predictions. The output layer uses a softmax function for multi-class classification:

$$\hat{y} = \text{softmax}(W_{\text{out}}\mathbf{h}_{\text{final}} + b_{\text{out}}) \quad (10)$$

where  $\mathbf{h}_{\text{final}}$  is the flattened attention output.

In Fig. 3, using TCNs and LSTMs together for IoV traffic takes use of their distinct strengths. TCNs (causal, dilated convolutions) get fine-grained, local patterns across long effective receptive fields with stable, parallelizable training. LSTMs, on the other hand, get order-sensitive, long-horizon dependencies with non-stationary context. Combining both and re-weighting important timesteps with multi-head attention creates more distinct temporal characteristics, which lowers false alarms and makes the system more stable against jitter and packet loss. In practice, the hybrid is good for edge deployment and provides quicker, more reliable federated convergence than either component alone. Algorithm 1 summarizes the procedural workflow described in this section. It trains each client's model by starting with the current global weights, fitting it to local data for  $E$  epochs, and producing updated local parameters. After training, it calculates and sends back the client's local weights, as well as the accuracy and loss, for aggregation and monitoring.

#### Algorithm 1 Local Model Training Process

**Input:** Rounds  $R$ , Client data  $\mathcal{D}_i = (X_i, y_i)$ , Global weights  $w_g$

**Output:** Local weights  $w_i$ , Performance metrics

**for each round**  $r = 1$  **to**  $R$  **do**

**for each client**  $i = 1$  **to**  $N$  **do**

        Initialize local model  $\mathcal{M}_i$  with weights  $w_g$ ;  
        Train  $\mathcal{M}_i$  on  $(X_i, y_i)$  for  $E$  epochs;  
        Obtain updated local weights  $w_i$ ;  
        Compute local accuracy and loss;

**return** local weights, accuracy, loss

#### D. Server Side Adaptive Aggregation Using Reinforcement Learning

A key issue in FL is the lack of flexibility of traditional static aggregation methods, such as FedAvg and FedProx, which use static or heuristic weights for the aggregation of client's updates irrespective of their individual contributions to the overall task across rounds. These conventional approaches are generally not capable of dynamically adapting to the

#### Algorithm 2 Global Model RL-Based Weights Aggregation

**Input:**  $\tau'$ : total federated rounds,  $w_i$ : local model weights,  $\mathcal{A}$ : local accuracies

**Output:**  $w^\tau$ : updated global model weights(GMU) initialization;

```

1 Initialize global model weights  $w^0$ ;
for each round  $\tau = 1$  to  $\tau'$  do
2   Form state vector:  $s_\tau \leftarrow \text{reshape}(\mathcal{A}_\tau)$ ;
3   Predict Q-values:  $Q_\tau \leftarrow \text{agent.predict}(s_\tau)$ ;
4   Compute Softmax weights:  $w_{RL}^\tau \leftarrow \text{softmax}(Q_\tau)$ ;
5    $w^\tau \leftarrow \sum_{i=1}^K w_i \cdot w_{RL}^\tau[i]$ ;
6   Set global model weights:  $w_g \leftarrow w^\tau$ ;
7   Evaluate  $w_g$  on test data  $(X_{\text{test}}, y_{\text{test}})$ ;
8   Compute metrics: Accuracy, ROC AUC, Precision, Recall, F1-Score, Loss;
9   Compute reward  $r_\tau$  from global accuracy;
10  Observe next state  $s_{\tau+1} \leftarrow \text{reshape}(\mathcal{A}_\tau)$ ;
11  Store transition:
    agent.remember( $s_\tau, \arg \max Q_\tau, r_\tau, s_{\tau+1}, \text{done}$ );
if agent.memory size  $>$  batch size then
12    agent.replay(batch size);
13  Save checkpoint:  $w_g \rightarrow \text{round}_\tau.\text{ckpt}$ ;
14  Track and store performance metrics in history arrays;

```

**return**  $w^\tau$  as final GMU

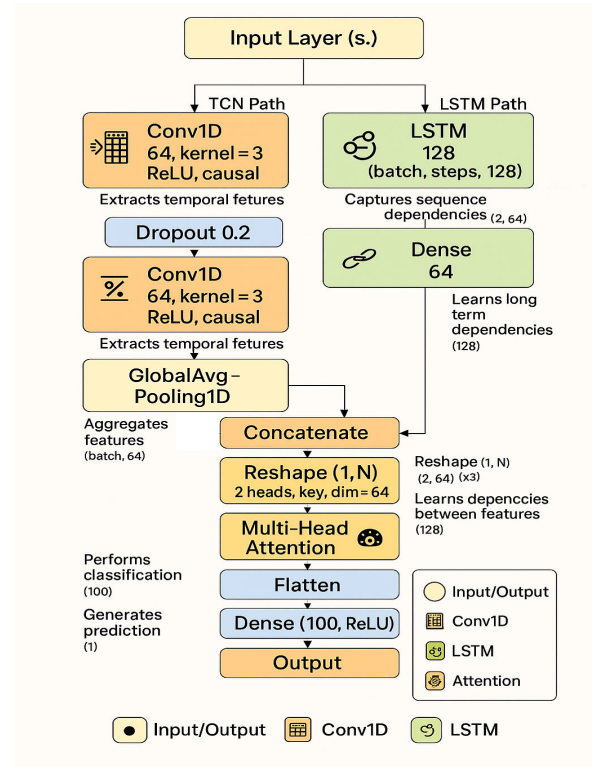


Fig. 3. Integrated temporal convolutional and LSTM network with attention mechanism.

cases of heterogeneous distributions of client data or changing local performance and thus, the global model accuracy is

suboptimal and convergence speed is slower. To overcome such limitation, in this paper, we propose a reinforcement learning-based adaptive aggregation strategy that allows for the dynamic optimization of the aggregation weights according to the real time feedback of the performance of client models.

In the proposed federated learning framework for secure IoV intrusion detection, the **server side adaptive aggregation** uses reinforcement learning (RL) to intelligently and dynamically combine the model updates from multiple clients. Once the locally trained model weights and associated performance metrics (e.g. accuracies or losses) are received from all participating clients, the central federated server will trigger the adaptive aggregation process. The server keeps these local weights and instead of static aggregation techniques such as FedAvg, uses a Deep Q-Network (DQN) agent to find the optimal aggregation weights for each client in each communication round.

In particular, having collected local accuracies (or any other utility metric of choice), the server creates a **state vector**  $s_t$  that encodes performance measures for the current round:

$$s_t = [a_1^t, a_2^t, \dots, a_K^t] \quad (11)$$

where  $a_k^t$  is the accuracy of client  $k$  in round  $t$ , and  $K$  is the number of clients.

The DQN agent receives this state vector and predicts a set of **Q-values** for each client, which are the expected future rewards (e.g., global accuracy improvement) when assigning different aggregation weights:

$$Q(s_t, a) = \mathbb{E}[R_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') \mid s_t, a] \quad (12)$$

Here,  $R_{t+1}$  is the immediate reward (global accuracy after aggregation),  $\gamma$  is the discount factor,  $a$  is an action (choices of weights of aggregation), and  $a'$  represents possible future actions.

Based on the predicted Q-values, the server uses a softmax function to transform these values to **adaptive aggregation weights** for each client:

$$w_k^t = \frac{\exp(Q(s_t, a_k))}{\sum_{j=1}^K \exp(Q(s_t, a_j))} \quad (13)$$

where  $w_k^t$  is the weight assigned to client  $k$  in round  $t$ .

The global model is then updated as a weighted average of local models:

$$\theta_{\text{global}}^t = \sum_{k=1}^K w_k^t \cdot \theta_k^t \quad (14)$$

where  $\theta_k^t$  are the local model parameters from client  $k$ .

After aggregation, global model is evaluated by the server on the validation set to compute the global accuracy which is used as the **reward** signal for the DQN agent. The experiences (state, action, reward, next state) are stored for replay and the DQN agent is periodically updated so as to enhance future aggregation policies. This cycle repeats in each round, enabling the system to continually adapt the importance given to each client based on observed contributions to global performance. For clarity and reproducibility, we summarize the DQN aggregator design below:

TABLE I  
CLASS DISTRIBUTION IN CICIOV2024 AND CICEVSE2024 DATASETS

| Dataset     | Class/Family   | Samples | Total   |
|-------------|----------------|---------|---------|
| CICIOV2024  | Benign         | 8,054   | 51,220  |
|             | DOS            | 8,378   |         |
|             | GAS            | 8,215   |         |
|             | RPM            | 8,068   |         |
|             | SPEED          | 9,041   |         |
|             | STEERING_WHEEL | 9,464   |         |
| CICEVSE2024 | Benign         | 14,364  | 115,299 |
|             | Attack         | 100,935 |         |

**State:**  $s_t = [a_t^k, l_t^k, n_t^k, \Delta a_t^k, \Delta l_t^k]_{k=1}^K$  (per-client accuracy, loss, normalized samples, and first-order deltas; size  $K \times d$ ). **Action/weights:** DQN outputs  $q_t \in \mathbb{R}^K$ ;  $w_t = \text{softmax}(q_t)$  with  $\sum_k w_t^k = 1$ ,  $w_t^k \geq 0$  (optional clipping  $w_{\min} = 0.02$ ,  $w_{\max} = 0.60$ ). **Reward:**  $r_{t+1} = \text{Acc}_{t+1} - \text{Acc}_t - \lambda \text{Var}_k(a_{t+1}^k) - \beta \|w_t - u\|_2^2$ ; (abl.:  $r_{t+1} = \text{Acc}_{t+1}$ ). **Network:** MLP (LayerNorm+ReLU,  $128 \rightarrow 64$ , output  $K$ ); target net, replay; Huber loss; Adam ( $1 \times 10^{-3}$ );  $\gamma = 0.95$ ;  $\varepsilon$ -greedy  $\varepsilon : 0.30 \rightarrow 0.05$ . Static global averaging utilizes fixed or size-based weights, whereas global server in proposed strategy uses a DQN (reinforcement-learning) approach to assign client weights adaptively each round based on objective performance feedback from held out assessments (such as accuracy or loss). Updates that are strong and dependable are given greater weight, whereas updates that are noisy or low-quality are given less weight. This makes the global model more stable when there are non-IID data, stragglers, and changes in distribution. This data driven weighting keeps convergence stable and speeds it up, lowers round-to-round fluctuations, and stops any one client with a skewed or uneven data distribution from dominating. In practice, this leads to better and more constant global accuracy and quicker recovery when conditions change, which is better than static global averaging. Algorithm 2 summarizes the procedural workflow described in this section. Construct a state from client accuracies, the reinforcement learning agent estimates Q-values, softmax-normalized into aggregation weights to update global parameters. Evaluate the global model, compute reward, perform experience replay, log metrics, and output the final GMU.

## IV. RESULTS AND DISCUSSIONS

### A. Dataset Collection

This study employs two established datasets, CICIOV2024<sup>1</sup> [9] and CICEVSE2024<sup>2</sup> [10], to comprehensively assess the proposed methodology. Table I shows the class distribution in both datasets. The CICIOV2024 dataset serves as an extensive benchmark for cybersecurity research in Internet of Vehicles (IoV) environments. It features a wide spectrum of cyberattack scenarios performed on a 2019 Ford vehicle, with all Electronic Control Units (ECUs) included. The dataset is organized into several attack categories, including Denial-of-Service

<sup>1</sup><https://www.unb.ca/cic/datasets/iov-dataset-2024.html>

<sup>2</sup><https://www.unb.ca/cic/datasets/evse-dataset-2024.html>

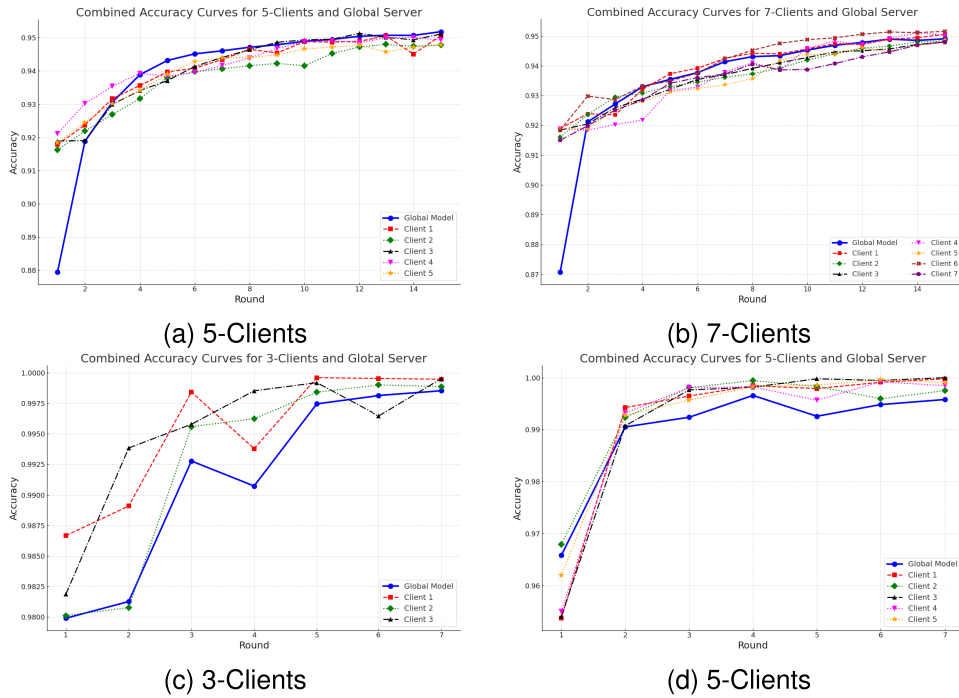


Fig. 4. Comparison of accuracy trends for various client counts and datasets: (a,b) CICEVSE2024 and (c,d) CICIOV2024.

(DoS), GAS, RPM, SPEED, and STEERING WHEEL, as well as benign samples. Specifically, CICIOV2024 contains 8,055 benign instances with multiple attack classes including 8,378 DoS samples, 8,215 GAS attack samples, 8,068 RPM attacks, 9,041 SPEED attacks and 9,464 STEERING WHEEL attacks. This gives a total of 51,221 data points. Each sample is given in binary, decimal, and hexadecimal formats and can be analyzed in detail and with great perspective. The CICEVSE2024 dataset focuses on the electric vehicle charging infrastructure and covers the benign and the attack scenarios. It reflects the vulnerabilities of the electric vehicle chargers in idle and active charging states, capturing various network and host-based attacks such as reconnaissance, DoS, backdoor, and cryptojacking. In terms of the class distribution, we have CICEVSE2024 which consists of 14,364 benign samples and 100,935 attack samples, for a total of 115,299 records. Prior to experimentation, both datasets go through rigorous pre processing in order to select only the most relevant and noise free features to be used for model development and evaluation. The dataset was then divided, 80% being used for the training and the remaining 20% being used for testing in the intrusion detection experiments.

### B. Performance Metrics

To have a thorough assessment of the proposed approach, we accounted for multiple rounds and also varied the number of clients. Precision, recall, F-measure, and accuracy were used as the evaluation measures. The metrics were calculated based on the number of True Positives (TP), False Positives (FP), True Negatives (TN) and False Negatives (FN). The resulting performance equations are given in Equations 15-18.

$$Precision = \frac{TP}{(TP + FP)} \quad (15)$$

$$Recall = \frac{FP}{(FP + TN)} \quad (16)$$

$$F - measure = \frac{(2 * TP)}{(2TP + FP + FN)} \quad (17)$$

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (18)$$

The proposed federated learning framework was created in PyCharm using Keras and evaluated on a computing environment with an NVIDIA GeForce RTX 2060 (8 GB) and 32 GB of RAM.

### C. Performance Analysis and Comparisons

The dynamic accuracy curves in Fig. 4 clearly demonstrate that the proposed strategy achieves rapid and stable accuracy improvements for both the global and client models in 5-client and 7-client scenarios using the CICEVSE2024 dataset. All models converge quickly and reach high accuracy values, highlighting the overall effectiveness of the approach for IoV intrusion detection. This pattern highlights the strong learning capacity and stability imparted by the temporal hybrid deep model, which leverages the complementary strengths of TCN and LSTM with attention mechanisms to capture both short-term and long-term dependencies in client data. The close alignment of client curves and the global model further indicates the balanced generalization across all participants and efficient knowledge integration through federated aggregation. The smooth and consistently high accuracy of the global model in both settings validates the adaptive reinforcement learning-based aggregation, which dynamically weighs and combines local updates from heterogeneous clients, thus minimizing divergence and ensuring robust global performance even as the number of clients increases. Notably, as more clients participate (in the 7-client case), the overall trend

TABLE II  
PERFORMANCE COMPARISON FOR DIFFERENT NUMBERS OF CLIENTS USING CICEVSE2024 DATASET

| Class  | 3 Clients     |            |              | 5 Clients     |            |              | 7 Clients     |            |              |
|--------|---------------|------------|--------------|---------------|------------|--------------|---------------|------------|--------------|
|        | Precision (%) | Recall (%) | F1-score (%) | Precision (%) | Recall (%) | F1-score (%) | Precision (%) | Recall (%) | F1-score (%) |
| BENIGN | 81.73         | 78.18      | 79.91        | 83.42         | 76.54      | 79.83        | 81.94         | 75.81      | 78.76        |
| Attack | 96.91         | 97.51      | 97.21        | 96.70         | 97.84      | 97.26        | 96.59         | 97.62      | 97.11        |

TABLE III  
PERFORMANCE COMPARISON FOR DIFFERENT NUMBERS OF CLIENTS USING CICIOV2024 DATASET

| Class          | 3 Clients     |            |              | 5 Clients     |            |              | 7 Clients     |            |              |
|----------------|---------------|------------|--------------|---------------|------------|--------------|---------------|------------|--------------|
|                | Precision (%) | Recall (%) | F1-score (%) | Precision (%) | Recall (%) | F1-score (%) | Precision (%) | Recall (%) | F1-score (%) |
| Benign         | 99.63         | 99.69      | 99.66        | 99.81         | 98.70      | 99.25        | 99.94         | 98.14      | 99.03        |
| DOS            | 99.88         | 99.76      | 99.82        | 99.64         | 99.58      | 99.61        | 99.40         | 99.40      | 99.40        |
| GAS            | 99.94         | 100.00     | 99.97        | 99.82         | 99.88      | 99.85        | 99.58         | 100.00     | 99.79        |
| RPM            | 99.75         | 99.88      | 99.81        | 99.26         | 99.88      | 99.57        | 99.20         | 100.00     | 99.60        |
| SPEED          | 99.89         | 100.00     | 99.94        | 99.56         | 99.94      | 99.75        | 99.67         | 99.89      | 99.78        |
| STEERING_WHEEL | 100.00        | 99.79      | 99.89        | 99.84         | 99.89      | 99.87        | 99.63         | 99.89      | 99.76        |

TABLE IV  
GLOBAL MODEL MAXIMUM AND MINIMUM ACCURACY (%) FOR DIFFERENT CLIENTS USING CICIOV2024 AND CICEVSE2024 DATASETS

| Dataset     | Clients | Maximum Accuracy (%) | Minimum Accuracy (%) |
|-------------|---------|----------------------|----------------------|
| CICIOV2024  | 3       | 99.85                | 97.99                |
|             | 5       | 99.66                | 96.58                |
|             | 7       | 99.57                | 93.17                |
| CICEVSE2024 | 3       | 95.10                | 87.49                |
|             | 5       | 95.18                | 87.94                |
|             | 7       | 94.90                | 87.07                |

of accuracy remains high, confirming the scalability and adaptability of the proposed methodology without sacrificing convergence speed or stability. Table II indicates that the proposed method achieves consistently high attack detection, with F1-scores of 97.21%, 97.26%, and 97.11% for 3, 5, and 7 clients, respectively. For the benign class, F1-scores are 79.91%, 79.83%, and 78.76% on the same settings. The results obtained show the good and robust intrusion detection performance of the framework, in particular for attack detection, even when the number of clients grows.

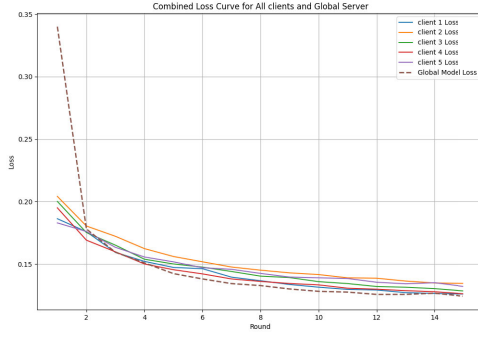
The dynamic accuracy curves Fig. 4 show the effectiveness of proposed strategy on CICIOV2024 dataset. For the 3-client and 5-client models, it is shown that all client models and the global model have very fast and stable convergence in the accuracy, reaching values above 99% in a short time. Comparing the curves of each client and the global model, we can see that the temporal hybrid deep model has a good ability to support resilient local training, and the adaptive reinforcement learning aggregation plays an optimal role in integrating client updates. This fact gives high and stable accuracy even if the number of clients is increased that shows the consistency and scalability of the approach. Table III also supports this finding, showing perfect precision, recall, and F1-scores across all attack families and for each client configuration. For example, in the 3-client case, benign traffic achieves a precision of 99.63%, recall of 99.69%, and F1-score of 99.66%, while DOS, GAS, RPM, SPEED, and STEERING WHEEL classes also report metrics above 99.8% in almost all cases. Similarly, in the 5-client and 7-client settings,

the results remain consistently high across all classes, with only minimal variations. These outstanding metrics confirm that the proposed framework can effectively and accurately detect both benign and attack classes across a diverse set of intrusion types, demonstrating not only strong convergence in training but also excellent generalization in distributed, real-world IoV environments. Even though client-level IID partitioning is applied, the benign class has a slightly smaller F1 score. This is primarily due to dataset-level class priors and feature similarity with low-intensity attacks, leading to more benign misclassifications. Simple, non-intrusive fixes like class-weighted loss, small benign oversampling in mini-batches, or post-hoc threshold calibration using ROC/PR can reduce this difference even smaller without changing the originally proposed federated methodology.

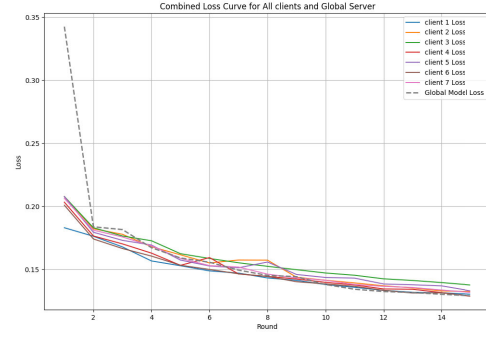
The loss curves in Fig. 5 clearly show that both the client and global model losses decrease rapidly and converge to low values as training progresses. This rapid and stable reduction demonstrates the effectiveness of the Temporal Hybrid Deep Model in extracting useful temporal features at each client, allowing for efficient local training. At the same time, the adaptive reinforcement learning aggregation consistently aligns the global loss curve with those of the clients, ensuring optimal global convergence and minimizing divergence. The close alignment between client and global loss curves highlights the strong collaboration and robust optimization achieved through this federated strategy. Table IV shows that the global model achieves high maximum accuracies across all client settings, with values ranging from 99.85% to 99.57%

TABLE V  
ROUND-WISE GLOBAL MODEL ADAPTIVE AGGREGATION PERFORMANCE (CICIOV2024, 3 CLIENTS)

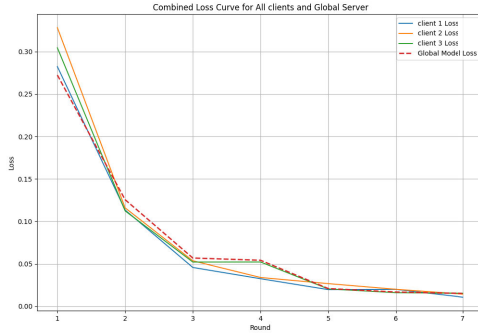
| Round | Accuracy (%) | ROC AUC (%) | Precision (%) | Recall (%) | F1-Score (%) |
|-------|--------------|-------------|---------------|------------|--------------|
| 1     | 97.99        | 99.91       | 97.95         | 97.95      | 97.93        |
| 2     | 98.13        | 99.95       | 98.10         | 98.07      | 98.05        |
| 3     | 99.28        | 99.99       | 99.26         | 99.26      | 99.25        |
| 4     | 99.07        | 99.99       | 99.05         | 99.06      | 99.05        |
| 5     | 99.75        | 100.00      | 99.75         | 99.74      | 99.74        |
| 6     | 99.81        | 100.00      | 99.81         | 99.81      | 99.81        |
| 7     | 99.85        | 100.00      | 99.85         | 99.85      | 99.85        |



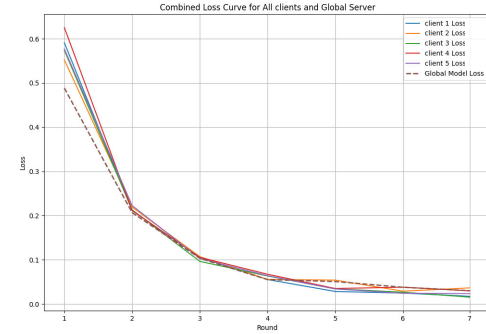
(a) 5-Clients



(b) 7-Clients



(c) 3-Clients



(d) 5-Clients

Fig. 5. Comparison of loss curves for various client configurations and datasets: (a,b) CICEVSE2024 and (c,d) CICIOV2024.

for CICIOV2024 and 95.18% to 94.90% for CICEVSE2024, while minimum accuracies remain above 93.17% and 87.07%, respectively. These results reflect the strong and consistent performance of the proposed framework in delivering reliable accuracy for intrusion detection, even as the number of clients increases. These findings show an optimal precision for curated datasets. However, we predict that accuracy will go downward somewhat under noisier real-world IoV conditions. This will be lessened by proposed DQN-driven aggregation, which automatically reduces the impact of unreliable client updates.

The results in Fig. 6, along with Table V and Table VI, show that RL based adaptive aggregation enables the global model to consistently match or exceed the accuracy of individual clients across rounds. This way, it guarantees speedy and stable improvements in the global model metrics like accuracy, precision, recall, and F1 score and it has shown a high degree of synchronization and successful integration

of client knowledge. The obtained results demonstrate the efficacy of the proposed scheme in providing high performance and reliable intrusion detection in various federated IoV environments. Fig. 6 illustrates the result of client's local model accuracy change curve against the federated rounds, compared to the global model accuracy. Fig. 6(a) shows results for the CICIOV2024 dataset, and Fig. 6(b) shows results for the CICEVSE2024 dataset with five participating clients. In both cases, the global model accuracy after the second round is very close to the client models and converges in the later rounds, demonstrating that the aggregation step is able to maintain client knowledge while not reducing the overall performance. Fig. 7 shows very high accuracy for intrusion detection by the proposed framework with very few misclassifications in all classes for the CICIOV2024 (3-clients) and CICEVSE2024 (5-clients) datasets. The true positive rate is high and the false positive as well as false negative is low in each confusion matrix, which indicates that the system is reliable and robust in

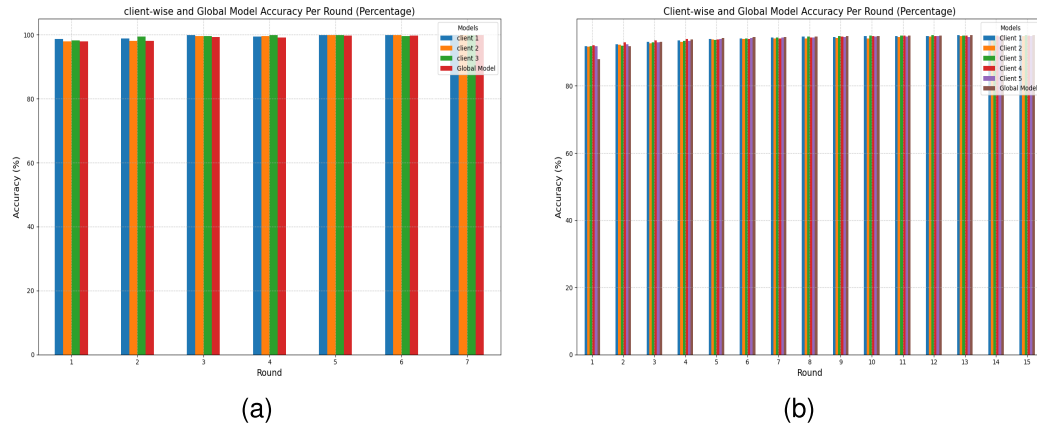


Fig. 6. Accuracy comparison of clients and aggregated global model per round for (a) CICIOV2024 and (b) CICEVSE2024 datasets.

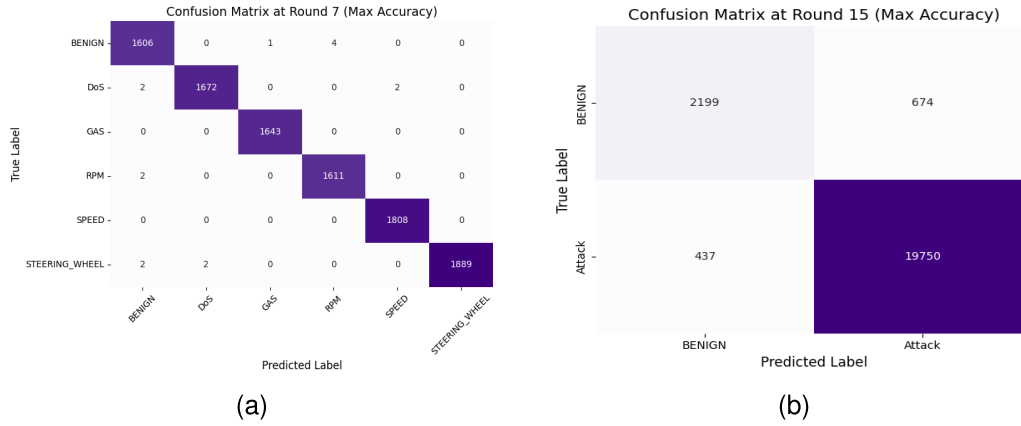


Fig. 7. Confusion matrix for intrusion detection using (a) CICIOV2024 dataset with 3-clients and (b) CICEVSE2024 dataset with 5-clients.

TABLE VI  
ROUND-WISE GLOBAL MODEL ADAPTIVE AGGREGATION PERFORMANCE (CICEVSE2024, 5 CLIENTS)

| Round | Accuracy (%) | ROC AUC (%) | Precision (%) | Recall (%) | F1-Score (%) |
|-------|--------------|-------------|---------------|------------|--------------|
| 1     | 87.94        | 89.77       | 90.40         | 96.47      | 93.34        |
| 2     | 91.88        | 95.45       | 91.67         | 99.79      | 95.56        |
| 3     | 93.06        | 96.24       | 93.60         | 98.84      | 96.14        |
| 4     | 93.89        | 96.68       | 95.09         | 98.08      | 96.56        |
| 5     | 94.31        | 96.92       | 95.96         | 97.61      | 96.78        |
| 6     | 94.51        | 97.10       | 96.36         | 97.41      | 96.88        |
| 7     | 94.61        | 97.25       | 96.50         | 97.37      | 96.93        |
| 8     | 94.71        | 97.34       | 96.68         | 97.31      | 96.99        |
| 9     | 94.79        | 97.41       | 96.79         | 97.28      | 97.03        |
| 10    | 94.89        | 97.48       | 96.72         | 97.47      | 97.09        |
| 11    | 94.95        | 97.50       | 96.75         | 97.50      | 97.13        |
| 12    | 95.04        | 97.56       | 96.62         | 97.76      | 97.19        |
| 13    | 95.07        | 97.59       | 96.77         | 97.63      | 97.20        |
| 14    | 95.07        | 97.53       | 96.71         | 97.69      | 97.20        |
| 15    | 95.18        | 97.61       | 96.70         | 97.84      | 97.26        |

both multi class and binary cases, and most of the predictions are along the diagonal of the confusion matrices, respectively. Fig. 8 and Fig. 9 clearly show that the IID data distribution scheme leads to higher and more stable accuracy for both the global model and individual clients compared to the non-IID setting. In Fig. 8, the IID curve rises sharply and converges smoothly to a higher accuracy, while the non-IID curve remains flatter with noticeable fluctuations and slower

improvement across rounds. Table VIII show that the IID data distribution scheme leads to higher and more stable accuracy for both the global model and individual clients compared to the non-IID setting in and Table VII. When averaged over 15 rounds, IID increases accuracy by 3.84 percentage points, ROC-AUC by 22.89 percentage points, precision by 5.52 percentage points, and F1-score by 1.93 percentage points. Recall stays extremely high in both settings. IID also

TABLE VII  
CLASSIFICATION PERFORMANCE WITHOUT IID BASED DATA DISTRIBUTION SCHEME (CICEVSE2024, 5 CLIENTS)

| Round | Accuracy (%) | ROC AUC (%) | Precision (%) | Recall (%) | F1-Score (%) |
|-------|--------------|-------------|---------------|------------|--------------|
| 1     | 87.54        | 65.28       | 87.54         | 100.00     | 93.36        |
| 2     | 87.54        | 66.69       | 87.54         | 100.00     | 93.36        |
| 3     | 87.54        | 50.00       | 87.54         | 100.00     | 93.36        |
| 4     | 87.54        | 66.60       | 87.54         | 100.00     | 93.36        |
| 5     | 87.54        | 73.07       | 87.54         | 100.00     | 93.36        |
| 6     | 87.54        | 92.85       | 87.54         | 100.00     | 93.36        |
| 7     | 91.05        | 72.32       | 90.72         | 100.00     | 95.14        |
| 8     | 91.09        | 75.78       | 90.76         | 100.00     | 95.16        |
| 9     | 91.50        | 70.47       | 91.16         | 99.99      | 95.37        |
| 10    | 91.06        | 82.98       | 90.74         | 100.00     | 95.14        |
| 11    | 91.88        | 76.01       | 91.57         | 99.93      | 95.57        |
| 12    | 91.86        | 75.37       | 91.58         | 99.89      | 95.55        |
| 13    | 92.88        | 75.34       | 92.96         | 99.39      | 96.07        |
| 14    | 93.23        | 78.99       | 93.37         | 99.32      | 96.25        |
| 15    | 92.50        | 84.29       | 92.33         | 99.71      | 95.88        |

TABLE VIII  
CLASSIFICATION PERFORMANCE WITH IID BASED DATA DISTRIBUTION SCHEME (CICEVSE2024, 5 CLIENTS)

| Round | Accuracy (%) | ROC AUC (%) | Precision (%) | Recall (%) | F1-Score (%) |
|-------|--------------|-------------|---------------|------------|--------------|
| 1     | 87.94        | 89.77       | 90.40         | 96.47      | 93.34        |
| 2     | 91.88        | 95.45       | 91.67         | 99.79      | 95.56        |
| 3     | 93.06        | 96.24       | 93.60         | 98.84      | 96.14        |
| 4     | 93.89        | 96.68       | 95.09         | 98.08      | 96.56        |
| 5     | 94.31        | 96.92       | 95.96         | 97.61      | 96.78        |
| 6     | 94.51        | 97.10       | 96.36         | 97.41      | 96.88        |
| 7     | 94.61        | 97.25       | 96.50         | 97.37      | 96.93        |
| 8     | 94.71        | 97.34       | 96.68         | 97.31      | 96.99        |
| 9     | 94.79        | 97.41       | 96.79         | 97.28      | 97.03        |
| 10    | 94.89        | 97.48       | 96.72         | 97.47      | 97.09        |
| 11    | 94.95        | 97.50       | 96.75         | 97.50      | 97.13        |
| 12    | 95.04        | 97.56       | 96.62         | 97.76      | 97.19        |
| 13    | 95.07        | 97.59       | 96.77         | 97.63      | 97.20        |
| 14    | 95.07        | 97.53       | 96.71         | 97.69      | 97.20        |
| 15    | 95.18        | 97.61       | 96.70         | 97.84      | 97.26        |

TABLE IX  
COMPARISON OF AGGREGATION METHODS FOR CLASSIFICATION PERFORMANCE (CICEVSE2024, 5 CLIENTS)

| Aggregation Method            | Accuracy (%) | ROC AUC (%) | Precision (%) | Recall (%) | F1-Score (%) |
|-------------------------------|--------------|-------------|---------------|------------|--------------|
| RL-based Adaptive Aggregation | 93.23        | 78.99       | 93.37         | 99.32      | 96.25        |
| FedProx Aggregation           | 92.28        | 74.76       | 92.05         | 99.80      | 95.76        |
| FedAvg Aggregation            | 92.48        | 75.13       | 92.32         | 99.71      | 95.87        |

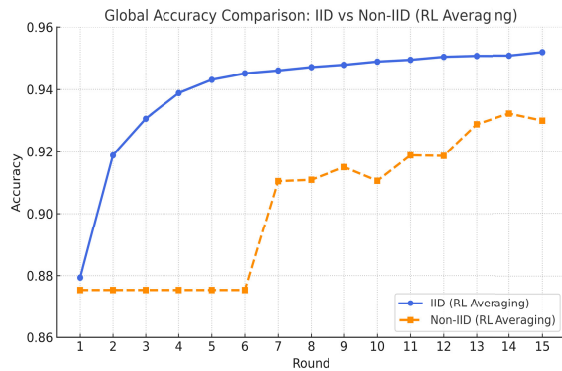


Fig. 8. Global model accuracy comparison: IID VS Non-IID data distribution scheme (CICEVSE2024 dataset, 5-clients).

converges more smoothly with less round to round fluctuation, which shows that balanced partitions lead to more stable

and discriminative federated learning performance. The bar graphs in Fig. 9 show that IID partitions provide increasingly greater and higher accuracies throughout rounds for all five clients. Non-IID partitions provide lower accuracies that exhibit more variability across clients and often achieve a plateau earlier. This ongoing gap between IID and Non-IID shows how data heterogeneity affects the stability of learning. Non-IID settings still have more round-to-round variation, even with Reinforcement-based aggregate. Adaptive weighting is helpful, but it doesn't completely make up for uneven client distributions. When IID settings are used, clients are more closely aligned, which means that generalization is stronger and more balanced. In general, balanced data allocation (with adaptive aggregation) speeds up convergence and enhances end-of-round performance in a Federated learning intrusion detection system. Fig. 10 and Table IX collectively show

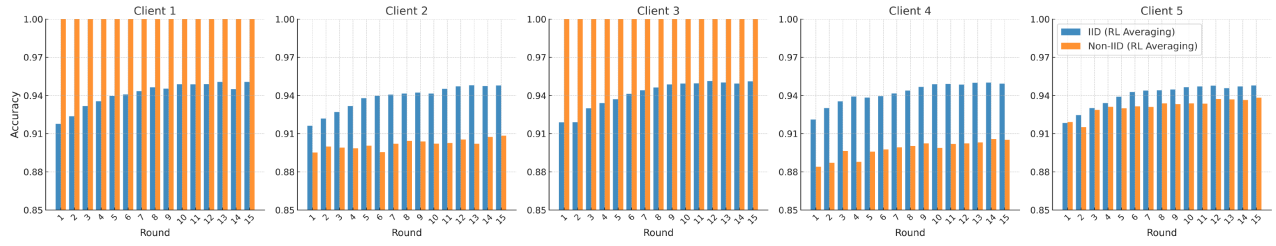


Fig. 9. Client wise accuracy comparison: IID VS Non-IID data distribution scheme (CICEVSE2024 dataset).

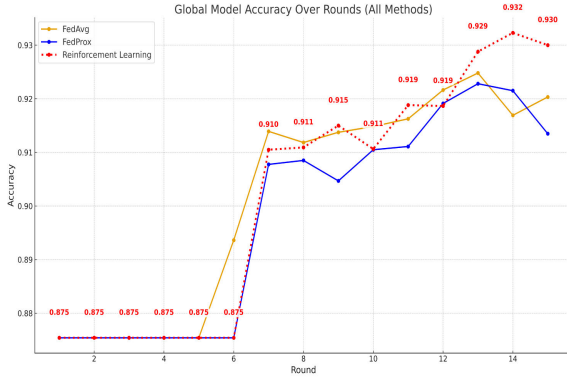


Fig. 10. Global model accuracy comparison for various aggregation methods (CICEVSE2024, 5- clients).

TABLE X  
PERFORMANCE COMPARISON OF THE PROPOSED MODEL WITH EXISTING TECHNIQUES

| Author & Year  | Accuracy (%) |
|----------------|--------------|
| FED-IDS [25]   | 99.72        |
| DL-IDS [26]    | 99.77        |
| DL-LeeNET [27] | 97.21        |
| Proposed Model | 99.85        |

that RL-based adaptive aggregation consistently outperforms both FedAvg and FedProx across all classification metrics using the CICEVSE2024 dataset with 5 clients. The RL-based method achieves the highest accuracy (93.23%), F1-score (96.25%), and strong precision and recall, while also delivering steady improvements throughout the rounds as seen in the figure. This demonstrates that reinforcement learning-based aggregation enables more effective and stable global model learning in federated environments compared to conventional aggregation approaches. To assess the effectiveness of the proposed scheme, its performance was benchmarked against established approaches, namely FED-IDS [25], DL-IDS [26], and DL-LeeNET [27]. Experimental results on vehicle-based IDS datasets demonstrate that the proposed model outperforms these existing methods. According to Table X, the proposed model achieves higher accuracy. Our approach has three clear benefits over current federated learning based IDS solutions. First, an adaptive aggregator based on DQN learns client weights for each round from performance feedback. The result renders the system more speed up when clients are not IID and stable convergence. Second, a temporal hybrid encoder (CNN-LSTM-Attention) models both short and long range dependencies in CAN and EVSE traffic for better intrusion

detection. Third, the design is ready for edge use with consumer IoV devices such as telematics and electric vehicle chargers.

## V. CONCLUSION

This paper presented an adaptive federated learning framework to address the security challenges faced by consumer electronics in Internet of Vehicles (IoV) environments, particularly targeting the limitations of traditional FL-based intrusion detection approaches with non-IID data and static aggregation methods like FedAvg and FedProx. By standardizing and distributing IoV network data IID-wise among clients, our method ensures consistent and privacy-preserving model training. Each client employs a Temporal Hybrid Deep Model (CNN-LSTM-Attention) initialized from the global model, enhancing the extraction of complex temporal and spatial patterns. A Deep Q-Network (DQN)-based Reinforcement Learning (RL) agent at the server dynamically determines aggregation weights based on client Q-values, adaptively optimizing the global model. Extensive experiments on CICIOV2024 and CICEVSE2024 dataset demonstrate significant gains, and the detection accuracy are 99.85% and 95.18%, respectively. The results show the robustness and flexibility of the proposed framework for IoV security in the real world. The research does not examine privacy primitives. Future work will explore how to handle clients who only participate a few times, which will make deployments even more reliable in the real world. It will also empower secure aggregation, differential privacy, and adversarial-robust training to make it more robust in real world deployments.

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